

ANEESH GROVER

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Experience

AcademicAlly

July 2024 – August 2025

Founding Developer – ACM Thapar Society

React, Python, Pandas, Openpyxl, BeautifulSoup

- Served **10,000+** students by developing a responsive timetable portal for fast, reliable class schedule lookup.
- Built ETL pipelines to transform and validate **50,000+** Excel rows into optimized JSON datasets, implementing custom error handling that reduced parsing failures by **95%**.
- Supported deployment and maintenance of the **live portal**, coordinating with team members.

Projects

FocusBoard - Whiteboard Task Capture System

Kotlin, FastAPI, CRAFT, TrOCR, Phi-3 Mini, TFLite

- Built the Android app in Kotlin with ML Kit document scanning and an experimental **on-device YOLOv26n/TFLite** whiteboard detector, chosen for mobile size constraints.
- Engineered a **3-stage OCR pipeline** using CRAFT for line segmentation, TrOCR for transcription, and Phi-3 Mini for structured calendar JSON generation from handwritten input.
- Designed per-device request queues to serialize uploads, preventing concurrent TrOCR worker spawning and **eliminating cross-device race conditions** under simultaneous load.
- Returned structured JSON to the app for user approval before committing tasks to the shared E Ink display.

Campus-Compute - Distributed Compute Framework

Python, FastAPI, WebSockets, Docker

- Architected a LAN-distributed compute system with a centralized coordinator and worker nodes, enabling automatic discovery and reconnect handling.
- Designed a capability-aware scheduler using CPU, RAM, and GPU metadata to optimize task placement across nodes.
- Built a real-time WebSocket dashboard for cluster telemetry, job orchestration, and containerized workload execution.

PatchIt - Civic Infrastructure Monitoring

PyTorch, YOLO, Depth-Anything-V2

- Co-led a crowdsourced platform with citizens and contractors to streamline reporting and resolution of road infrastructure issues.
- Built a YOLO-based pothole detection pipeline achieving **85.8% precision** and **81.7% mAP@50**, integrating depth estimation for 3D geometry and repair cost estimation.
- Runner-up (Open Innovation) at **NVIDIA SatHack Hackathon** among 1500+ participants.

DeepResearch - Multi-Agent Research Assistant

Streamlit, FAISS, HuggingFace, LLaMA3

- Built a multi-agent research system using LangGraph, integrating search APIs and LLMs for automated synthesis of structured, citation-backed insights.
- Implemented FAISS-based retrieval with embedding-driven indexing for context-aware query resolution.
- Designed a citation-aware RAG pipeline ensuring traceable outputs grounded in reliable sources.

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Software Engineering & Systems: REST APIs, distributed systems, asynchronous processing, WebSockets, system design

Machine Learning: PyTorch, TensorFlow, transformers, vector databases, RAG

Infrastructure & Tools: Docker, Git, CI/CD, Supabase, OpenCV, CUDA

Education

University of Toledo

Bachelor of Science in Computer Science and Engineering

Toledo, OH

GPA: 3.5/4.0

Leadership & Involvement

Association for Computing Machinery (ACM)

Nov 2023 – Present

Executive Board Member & Machine Learning Bootcamp Lead

- * Organized Hacklipse 2024, ACM Thapar's flagship hackathon, coordinating operations for **300+ participants**.
- * Mentored **100+** students through ACM's Machine Learning Bootcamp by designing curriculum and leading weekly technical sessions.

Achievements

- Secured **runner-up** in **Open Innovation Category** at **NVIDIA's SatHack**, competing against **1500+** participants.
- Awarded Best Use Case for AcademicAlly at ACM Projectathon 2024.